Ex No. :- 3a

Date :- 23/07/26

UNDERSTANDING THE CISCO PACKET TRACER ENVIRONMENT

AIM :-

To understand the Cisco Packet Tracker environment and design a simple computer network using its interface and networking components.

STEPS :-

- Open the CISCO Packet Tracer application.
- Observe the user interface, including the menu Bar, Main Toolbar, Network Component Box and Real time / Simulation tabs.
- From the Network Component Box, select the required network devices such as PCs, Hub and switch.
- Drag and Drop the devices into the workspace.
- Connect the devices using copper straight through cables.
- Configure IP address
- Verify that all links are active
- Use the add simple PDU Tool to send a packet between two PC's
- Observe Packet transmission in Real time Mode and simulation Mode.
- Save the Network Topology

Date
Page

RESULT :-

The Cisco Packet Tracer environment was successfully explored and a simple network was designed. The interface components, devices connections, IP configuration & Packet transmission were successfully understood.

Exc: 3.b.1
Date: 23/07/26

HUB

AIM:-

To study the behaviour of a Hub by observing how it forwards data Packets using Cisco Packet Tracer.

STEPS:-

Open CISCO Packet Tracer

Drag and Place, Hub & 4 Generic PC's into the workspace

Connect all PC's to the Hub using Copper straight - through Cables.

Configure IP address for all PC's in same network.

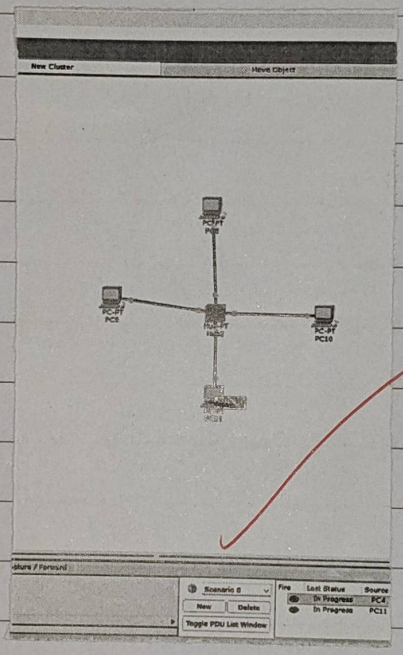
Add simple PDU from common tools bar.

Send a Packet from one Pc to another PC.

Switch to simulation Mode and Observe how the hub forwards the Packet.

RESULT:-

The hub broadcasts the received packet to all connected devices. Every PC receives the packet, but only the destination PC accepts it, while the other discard it.


New Cluster
Move Object
PC-PT
PC9
PC-PT
PC8
PC-PT
PC10
Scenario 0
New
Delete
Toggle PDU List Window
Fire
Last Status
Source
In Progress
PC8
In Progress
PC11

Exp: 3b-2

Date: 23/07/26

SWITCH

AIM :-

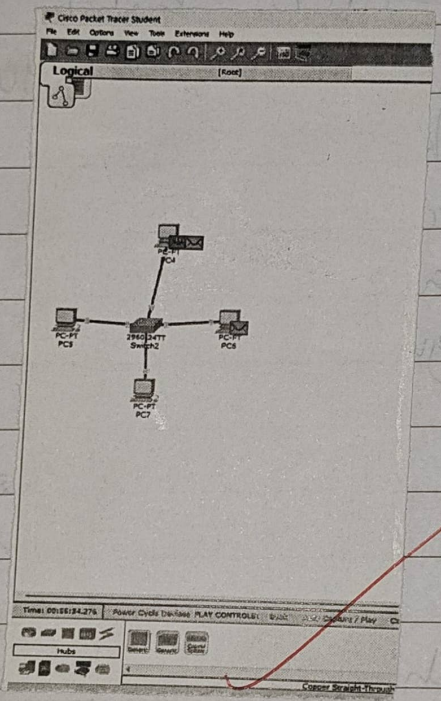
To Study the behaviour of a switch by observing how it forwards data packets using CISCO Packet Tracer.

STEPS :-

- Open CISCO Packet Tracer
- Drag and Place 1 switch and 4 generic PC's into the workspace.
- Connect all PC's to the switch using Copper straight - through Cables.
- Configure IP address for all PC's in the same network
- Select add simple PPU from the Common tools bar
- Send a packet from one PC to another PC
- Switch to simulation mode & Observe.

RESULT :-

The switch forwards the Packet only to the destination PC by using its MAC address table. This reduces unnecessary network traffic and improves communication efficiency.


Cisco Packet Tracer Student
File Edit Options View Tools Extensions Help
Logical
[Root]
PC-PT
PC4
PC-PT
PC3
2960-24TT
Switch2
PC-PT
PC6
PC-PT
PC7
Time: 00:16:34.276
Power Cycle Devices
FAST FORWARD / Play
Hubs
Copper Straight-Through

Exp: 3b3
Date: 23/07/26

HUB AND SWITCH COMPARISON

AIM :-

To compare the packet forwarding behaviour of a Hub and a switch using CISCO packet tracer

STEPS :-

Create one network using a Hub and another using a switch, each connected to 4 PC's

Configure IP addresses for all PC's in both networks

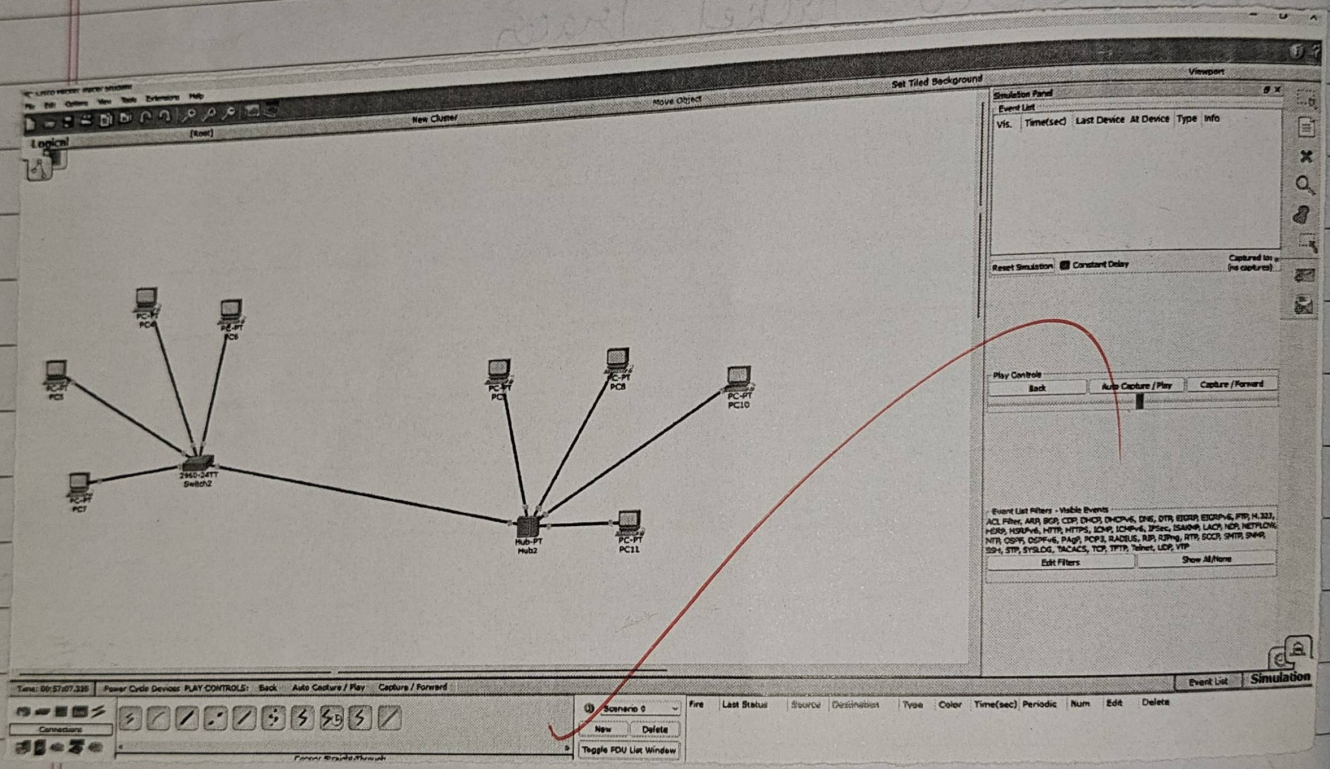
Send a simple PDU between Two PC's in each network.

Observe Packet Transmission in simulation mode.

Compare how the Hub & switch forward Packets.

RESULT :-

The Hub broadcasts Packets to all connected devices, whereas the switch forwards packets only to the intended destination. Therefore, the switch provides better performance, reduced network traffic, and improved efficiency compared to the HUB.


Logical
New Cluster
Move Object
Set Tiled Background
Viewport
Simulation Panel
Event List
Vis.
Time(sec)
Last Device
At Device
Type
Info
Reset Simulation
Constant Delay
Play Controls
Back
Auto Capture / Play
Capture / Forward
Edit Filters
Show All/None
PC-PT PC4
PC-PT PC5
PC-PT PC6
PC-PT PC7
2960-24TT Switch2
PC-PT PC8
PC-PT PC9
PC-PT PC10
Hub-PT Hub2
PC-PT PC11
Event List
Simulation
Scenario 0
New
Delete
Toggle PDU List Window
Fire
Last Status
Source
Destination
Type
Color
Time(sec)
Periodic
Num
Edit
Delete